

Monday Warm Up: Unit 4, Self-inductance, and RL circuits

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1 Memory Bank

- $\epsilon = -N\Delta\phi_m/\Delta t$... Faraday's Law
- $\epsilon = -L\Delta I/\Delta t$... Faraday's Law
- $\phi_m = \vec{B} \cdot \vec{A} = BA \cos(\theta)$... Definition of magnetic flux
- $\vec{B} = \mu_0 n I \hat{k}$... Magnetic field in a solenoid of length L , turns N , current I , and $n = N/L$.
- $1 \text{ H} = 1 \Omega \text{ s}$... The *Henry* is a unit of inductance, often represented by L .
- $\mu_0 = 4\pi \times 10^{-7}$... $\text{T m}^{-1} \text{ A}^{-1}$.
- $I(t) = I_0 (1 - \exp(-t/\tau))$... Current when RL circuits turn on ($\tau = L/R$).
- $I(t) = I_0 \exp(-t/\tau)$... Current when RL circuits turn off ($\tau = L/R$).

2 Self-Inductance

1. Suppose a current $I(t)$ is passed to an inductor. At $t = 0$ seconds, $I = 0$. The current increases linearly to 100 mA in 0.5 seconds, then remains constant for 2 seconds. At that point, it decreases to 0 mA in 0.5 seconds. (a) Sketch the current below. (b) Sketch the back emf across the inductor below. What are the values of the back emf during this process?

3. Using the result from the previous problem, (a) show that the inductance of a solenoid with length l , area A , and turns N is

$$L = \mu_0 AN^2/l \quad (1)$$

- (b) What is the inductance of a solenoid with area $A = 0.1 \text{ cm}^2$, $l = 2 \text{ cm}$, and turns $N = 100$?

3 RL Circuits

1. (a) What is the characteristic time constant for a 9.0 mH inductor in series with a 3.00Ω resistor? (b) Find the current 6.00 ms after the circuit is closed. (c) If the circuit is powered by 5.0 V, what is the current as $t \rightarrow \infty$?

2. Derive the general relationship between inductance, L , and current, by equating the induced emf in the two versions of Faraday's Law.